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User-Defined Gesture Control System for Photography Drones

by

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Thesis presented in partial fulfilment of the requirements for the degree of Master of Engineering (Mechanical) in the Faculty of Engineering at Stellenbosch University

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May 2026

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Abstract

User-Defined Gesture Control System for Photography Drones

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May 2026

Drones are increasingly used in fields such as photography, inspection, and emergency response, yet their operation still requires specialist training and conventional control interfaces that can be difficult for non-experts to use. This project investigates a gesture-controlled drone system that allows users to define and personalise their own hand gestures for controlling a drone in a more natural way.

The proposed system will focus on learning user-specific gestures rather than forcing all users to adapt to a fixed gesture vocabulary. To support this, the project will explore few-shot learning methods for recognising new gestures from only a small number of examples, while retaining a small set of fixed safety-critical commands such as take-off, landing, return home, and emergency stop. The prototype will be implemented and evaluated on a physical drone to assess whether personalised gesture learning can provide reliable control for both basic flight and photography-oriented functions.

By combining gesture recognition, incremental learning, and real-world drone testing, the project aims to contribute to Human-Robot Interaction research and to demonstrate a more accessible and adaptable approach to drone control.

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Chapter 1

Introduction

1.1 Background

Drones, or Unmanned Aerial Vehicles (UAVs), are becoming increasingly important across many industries. They show great promise in simplifying logistics, reducing costs, improving human safety, and providing faster response times and more accurate results (Patrona *et al.*, 2021). Their effectiveness has been demonstrated across a wide variety of sectors, including agriculture, search and rescue, surveillance, transportation, and photography (G R, Divya *et al.*, 2025). Despite these advantages, effective drone operation currently requires specialist training, limiting accessibility for many potential users (Int, 2017).

This has created demand for systems that allow users to control drones in a more natural and intuitive way. The field of Human-Robot Interaction (HRI) focuses on the study and design of communication between humans and robots (ScienceDirect, 2026), and has driven the emergence of Natural User Interfaces (NUIs). NUIs encompass a range of technologies, including touchscreens, voice recognition, eye tracking, and gesture recognition (RTI International, 2023). This project will focus on gesture control as a natural interface for operating a photography drone.

Gesture control enables users to communicate with digital systems through natural hand and body movements, removing the need for traditional input devices (Anuja Phapale and Shriya Sawashe, 2024). Research has shown that, without prior instruction, humans naturally attempt to communicate with drones through gestures (Gio *et al.*, 2021), suggesting that gesture-based control aligns closely with instinctive human behaviour. This has practical implications as gesture-based drone control lowers the barrier to entry to drones by removing the need for specialist training (Mehta *et al.*, 2023). Additionallly, gesture control has been shown to enable both experienced and novice pilots to operate drones more effectively (Gio *et al.*, 2021).

Despite these advances, current gesture-based control systems remain limited in their flexibility. Most existing approaches rely on fixed, predefined gesture vocabularies that all users must conform to, with no accommodation for individual differences in movement style or preference (Reyes *et al.*, 2023). This lack of personalisation represents a clear gap in the research, which this project aims to address.

1.2 Aim

The aim of this project is to develop a gesture-controlled photography and videography drone that allows users to create and personalise their own gestures. Rather than conforming to a fixed set of predefined gestures, the system will learn and adapt to gestures chosen by the user, giving them full control over how they will interact with the drone.

A small set of core gestures will remain fixed for essential safety-critical functions such as take-off, landing, emergency stop, and return home. All other functions will be fully customisable, allowing the user to assign any gesture they choose to a specific task.

The drone will support a range of operations, from basic flight controls such as movement and rotation, to more complex photography-specific movements including tracking shots, dronies[1], spotlight shots, and reveal shots.

1.3 Objectives

In order to achieve this aim, the following objectives must be met:

- Implement system on a physical drone.
- Recognise and interpret basic gestures.
- Control basic drone functions using basic gestures.
- Learn unique user-defined gestures and assign them to photography-specific functions.
- Perform photography-specific functions.
- Manage similar or ambiguous gestures reliably.

1.4 Motivation

Gesture-based drone control presents complelling alternative to traditional contollers. By leveraging the natural human motor skills of a person, it allows users to interact with a drone in a more natral way, reducing the complxity assosiated with conventional control interfaces (Trinitatova *et al.*, 2025). As a result this will greatly improve the accessabiliy as users without prior piloting experience. Taylor *et al.* (2025) has stated that people without prior piloting exerience have been shown to operate gesture-controlled drones safely and intuitively, particulary when combined

[1] A dronie is a cinematic shot where the drone moves backward and upward while keeping the subject centred in the frame

with features such as obstacle avoidance and 3D trajectory mapping (Taylor *et al.*, 2025).

Beyond accessibility, gesture control can reduce cognitive load by enabling more instantaneous and instinctive command inputs, which is particularly valuable in time-critical applications such as search and rescue or disaster relief scenarios (Kumar *et al.*, 2020). Lightweight, real-time gesture recognition algorithms have further demonstrated that a manageable set of gestures can support precise and nuanced drone manoeuvres without overwhelming the user (Yun *et al.*, 2024). Incorporating personalisation takes this further, allowing the intuitive nature of gesture interaction to be fully realised for each individual user.

1.5 Scope

The primary focus of this project is the gesture recognition system. Specifically, how accurately the system can recognise, classify, and learn new gestures. To achieve this, Few-Shot Learning (FSL) will be used, enabling the system to recognise a new gesture from only a small number of examples. This is essential for a system designed to learn from the user in real time.

A photography drone was chosen as the platform for testing because it offers a wide and complex range of tasks, making it an ideal testbed. Although the underlying system could be applied to many other domains; such as agriculture, search and rescue, or surveillance; the photography use case was selected as it presents the greatest variety of distinct functions. If the system performs effectively in this context, it is logical to assume that it wil also be effective in other industries.

The key gaps this project addresses are:

- The lack of truly personalised gesture control, where the user rather than the designer defines the gesture vocabulary.
- The challenge of incremental gesture learning — how a system can be introduced to a new gesture, correctly classify it, and reliably distinguish it from similar gestures going forward.

Chapter 2

Literature review

This chapter will discuss the current state of the art technology that incorporates gesture recognition to control the movement of a drone as well as gesture recognition as a whole. This chapter will discuss the various forms of gesture control, the learning algorithems behind gesture recognition and categorisation, and the current implementations of gesture control for drones.

2.1 Types of Gesture Control

2.1.1 Image-Based vs Non-Image-Based Gesture Recognition

According to Khaksar *et al.* (2023) there are two general approaches when it comes to gesture recognition and data acquisition: image-based and non-image-based approaches. The image-based approach can be further categorised into three main subcategories which include depth camera detection, stereo camera detection and single camera detection as shown in Figure 2.1 (Khaksar *et al.*, 2023). This approach makes use of cameras to capture a gesture and then using feature extraction, transforms this information into usable data that a system can use.

The non-image based approach on the other hand makes use of sensors to capture a gesture. These sensors are catogorised into glove devices and band devices. Typically these devices make use of Inertial Measurment Units (IMUs) in combination with flex sensors to capture the movement of a hand or arm (Müezzinoğlu and Karaköse, 2021). This method typically more robust to environmenteal factors (such as lighting conditions) compared to image-based approachs, however the recent trend has been to move to image-based approachces as they are less intrusive and more userfriendly (Gio *et al.*, 2021).

2.1.2 Gesture Languages

When a gesture is captured using either vision-based methods or non-vision based methods, movements of the hand are captured, however, theses movemnts need to be interpreted to give them meaning. For example, if a user moves their hand to the left, this movement needs to be interpreted as a command for the drone to move left. In order to do this, a gesture language is needed. Simply put, a gesture language is a dataset that matches a gesture to a specific meaning (Patrona *et al.*, 2021).

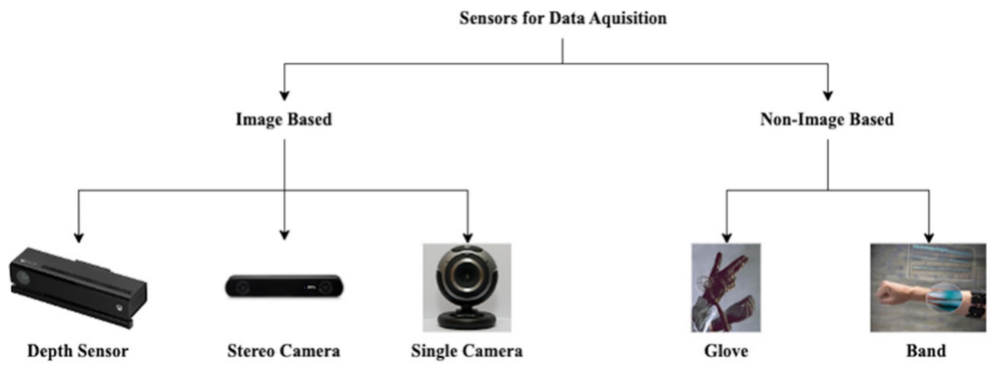


Figure 2.1: Flowchart showing the different types of gesture recognition approaches.

There are a numebr of different languages that have been used in the past for gesture recognition and have commonly been used to communicate between manned aerial vehicles. Patrona *et al.* (2021) expalins that aeroplane pilotes and helicopter pilots have been using hand gestures to communicate with each other for a number of years.

These gestures have also been used to communicate with other unmanned vehicles, most notably drones. Patrona *et al.* (2021) lists the some of the most common gesture languages used to communicate with arial vehicles:

- **Naval Air Training and Operating Procedures Standardization (NATOPS).** This is the most complete and detailed dataset, used for communicating between a pilot of a aircraft or helicopter and the ground crew. This meanis that it cannot be used for drone control as "The way UAVs "perceive" what they see is totally different from the way humans do so." (Patrona *et al.*, 2021).
- **DJI Spark Language**. This lanugae was launched in 2017 as the first commerciral UAV to have gesture control for handeling the movement of the drone as well as image/video capturing. It consists of 8 gestures, namely: "palm launch, palm control, adjusting position, stop adjusting position, follow, take selfie, record video, beckon and palm land"
- **DJI Mavic Air Lanhuage.** This language is similar to the DJI Spark language, however with a few alternations and additions to the gestures.
- **Keck Gesture Dataset.** This is a subset of the NATOPS language and it consists of 14 gestures that were designd to be used in a cluttered environment with moving cameras. Its is primeraley used to validate other gesture languages as it is a framework with 126 videos of static gesrtures and 168 videos of gesutuses in a dynamic environment (Escalera, Sergio, 2017).
- **UAV-GESTURE Dataset.** This dataset was created in 2018 and was the first

publically available dataset of "UAV handling signals recoprded outdoors by a flying drone." (Patrona *et al.*, 2021).

- **AUTH UAV Gesture LAnguage**. This dataset was created by Patrona *et al.* (2021) as a more generic language for controlling a dataset that improved on the limitations of the previously mentioned datasets.

Although these gesture languages and datasets are effective for drone control, they are limited by the need for users to learn application-specific gestures. This restricts the ability to personalise the interaction, meaning users cannot easily employ natural gestures or tailor the system to suit their own requirements. The aim of this project is therefore to bridge this gap by allowing users to design and customise their own drone control gestures.

2.2 Current Implementations

There have been a numebr of implementations of gesture control for drones. They range in their application, harware implementations as well as software implementations. A few will be discussed in this report

2.2.1 Wearable Glove Control

One of the most comman implementations of gesture controlled drones has been to use a wearable device. The basic idea is that the drone mimics the movements of the hand and incorporates additional sensors to capture a wide range of gestures. A study by Müezzinoğlu and Karaköse (2021) proposed an intelligent human-UAV interaction framework that was cenetered around the use of two wearable gloves that allowed for real-time, controller-free piloting of a drone.

Each glove is made up of 5 flex sensors that are used to capture the bending movemnts of the fingers. These sensors are made of a resistive material that changes its resistance when bent. As the resistance changes an analog signal is created and can be processed to determine the shape of the hand. The gloves also make use of an IMU incorporating a 3-axis gyroscope and 3-axis accelerometer. These sensors work together to capture the orientation and movement of the hand. The data is then captured using an STM32 microcontroller.

Figure 2.2 depicts the design of the glove (Müezzinoğlu and Karaköse, 2021).

As the user moves their hands and creates gestures, the sensors capture these signals. These signals are pre-processed by trimming and normalising the data. The signal triming phase is used to remove the first and last 3 % of the signal data to eliminate any electrical noise as well as the noise created by transitioning between different gestures. The normalisation phase is used to normalise the raw analog-to-digital

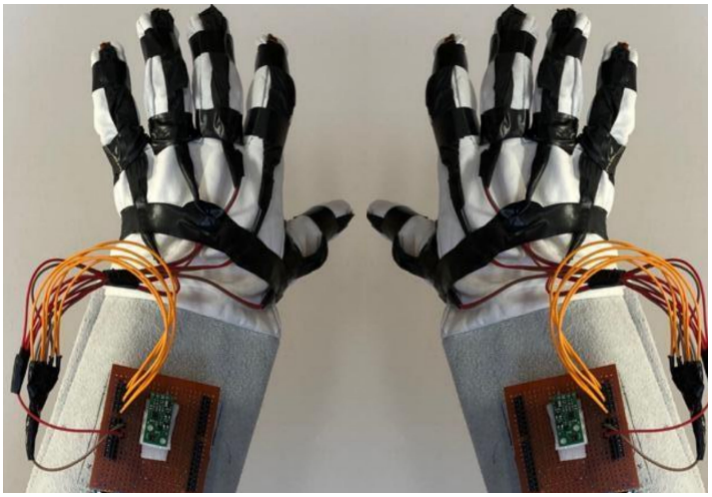


Figure 2.2: Design of the wearable glove used for gesture control of a drone.

values into a consistent dataset to ensure that different users produce recognisable data patterns.

Following this pre-processing, the framework employs a machine learning inference to classify the processed data into one of 25 distinct hand gestures.The researchers evaluated four specific machine learning algorithms to determine the most effective model for real-time flight control: Decision Tree (DT), Naive Bayes (NB), Support Vector Machine (SVM), and K-Nearest Neighbors (KNN). The experimental results revealed critical trade-offs between classification accuracy and computational speed. SVM was found to be the most precise model, achieving a mean accuracy of 98.02%, though it required the longest training time at approximately 20 666.7 ms. For real-time piloting where low latency is paramount, KNN proved superior, delivering the fastest test time of 1.22 ms. Decision Tree offered the most efficient training phase at 133.1 ms, while maintaining a competitive accuracy of 96.96%.

Once a gesture is recognized, the system utilizes a task scheduling algorithm to translate the identified class into pre-defined UAV flight commands. This multi-mode command structure allows for complex 3D navigation; for instance, specific hand orientations and finger flexions can trigger simultaneous movements such as “Forward Up” or “Turn Left Down”. This glove-based approach offers a significant engineering advantage over vision-based systems, as it remains robust against variable ambient lighting and line-of-sight obstructions that typically hinder camera-reliant gesture recognition. The system’s high accuracy and millisecond-level response times demonstrate its viability for high-precision human-UAV interaction in complex environments. One downside to this design is that it requires the user to wear a glove, making the device "intrusive and constraining for the user" (Gio *et al.*, 2021).

2.2.2 Skywriter HAT sensor

Mehta *et al.* (2023) made use of a near-field elctric field sensor (Skywriter HAT sensor shown in Figure 2.3 (Pimoroni, 2026)) to control the roll, pitch and yaw of a drone. The sensor detects a change in a magnetic field when a conducticve object (such as a hand) enters in the field. The change in magnetic field is converted into a digital signal that represents the x, y and z coordinate values. These values are used by a Raspberry Pi that converts these values into the roll, pitch and yaw control signals of a Crazyflie drone. An Extended Kalmen Filter was used to smooth the signals noise and to combine the information into one reliable estimate of the drone's position, velocity, and orientation at any moment.

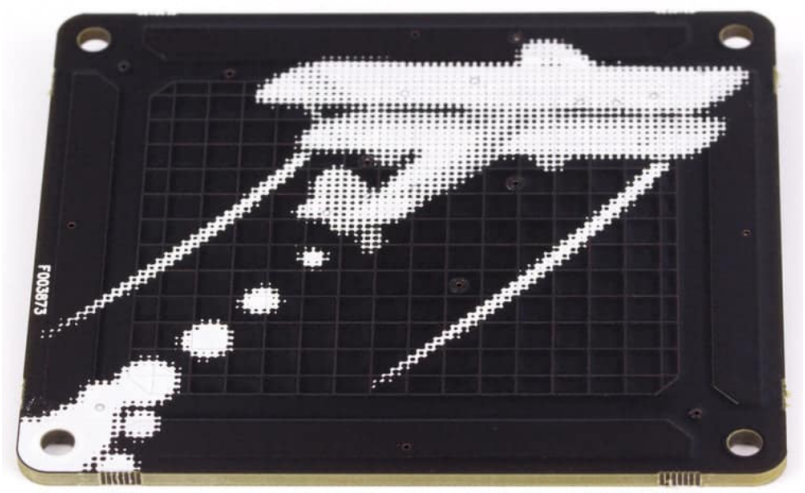


Figure 2.3: Skywriter HAT sensor used for gesture control of a drone.

The authors reported some success with this design; however, it also presented notable limitations and concerns. In particular, the Skywriter sensor was noisy, and the researchers were unable to fully isolate which axis was being moved. As a result, the drone drifted even when the user's hand remained stationary.

2.2.3 LEAP Motion Sensor

Sarkar *et al.* (2016) describes the Leap Montion Sensor as a gesture recognition device that uses two monochromatic infrared (IR) cameras and three infrared LEDs to track infrared light. By using stereoscopy from its two cameras, the Leap reduces errors and achieves highly precise tracking of hand and finger movements. The sensor can track hand motion up to a distance of about 1 meter directly above it, creating a hemispherical area above the sensor. Once the hand enters the area, the sensors detects movement and using the LEAP Software Development Kit (SDK), advanced Computer Vision algorithems run in order to exctract raw sensor data.

Figure 2.4 (Sarkar *et al.*, 2016) shows the motion sensor as well as the area where a hand movement can be detected.

Figure 2.4: Leap motion sensor

The Leap motion sensor has been used by Sarkar *et al.* (2016) as a ground station to control the movements of an AR DRONE 2.0. However this sensor was only used to control the pitch, yaw and roll of the drone and it was not used to communicate any higher level commands to the drone.

2.2.4 Microsoft Kinect

Gio *et al.* (2021) developed a system that uses the Microsoft Kinect to record full body gestures that were used to control a drone. The Microsoft Kinect is a depth-sensing device that captures the full body movements by constructing a real-time 3D skeletal model of a user. This model tracts the positions of the users jouints (eg. shoulders, elbows, spine and hips) which allows the system to track the posture and movement of a user. The gesture recognition is done by computing the angles between the joints rather than relying on the absolute positions of the body parts (e.g. arms, shoulders and hands). These calculated angles are compared to experimentally defined thresholds to determine which gesture is being peformed. This allows for gestures to be interperated, regardless of the users's body size or the distance from the sensor, making the system more felxible for users. Figure 2.5 shows a flow chart of how the system interacts with each subsystem (Gio *et al.*, 2021).

The Kinect streams the joint position data to a computer, where the motion recognition software evaluates the geometroc relationships between the various joints. These relationships and angles correspond to drone commands such as move forward, sideways or rotate. Figure 2.6 shows the commands and the associated gestures that Gio *et al.* (2021) has developed.

One key contribution of this work is its implementation of speed control, which is achieved using the amplitude of the user's movements. Instead of gestures acting a simple on/off commands, the magnitude of the body motion is mapped to the

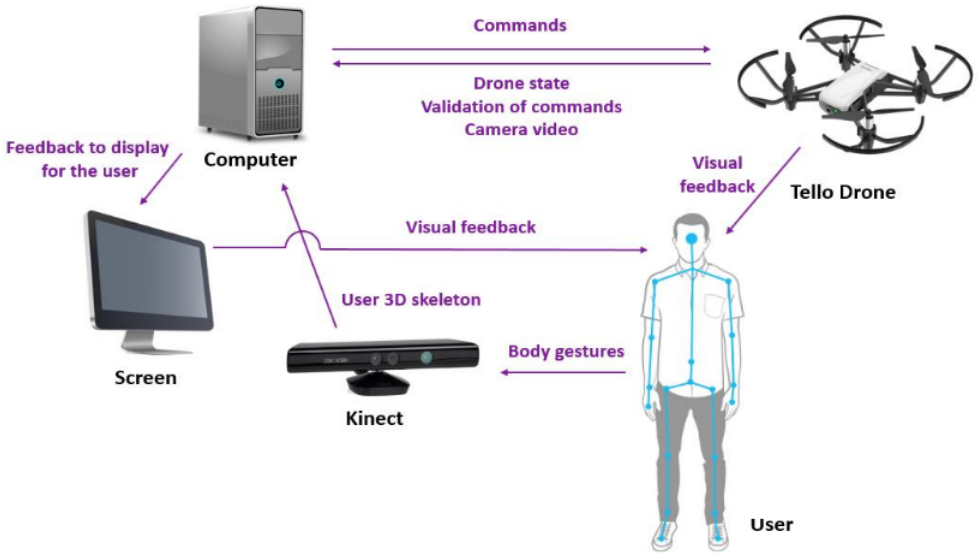


Figure 2.5: Overview of the Microsoft Kinect system for drone control

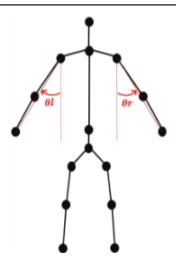
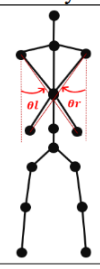
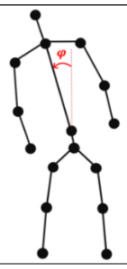
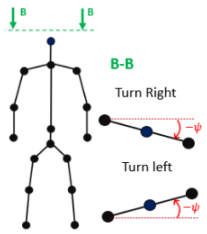
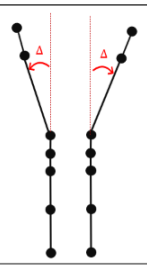
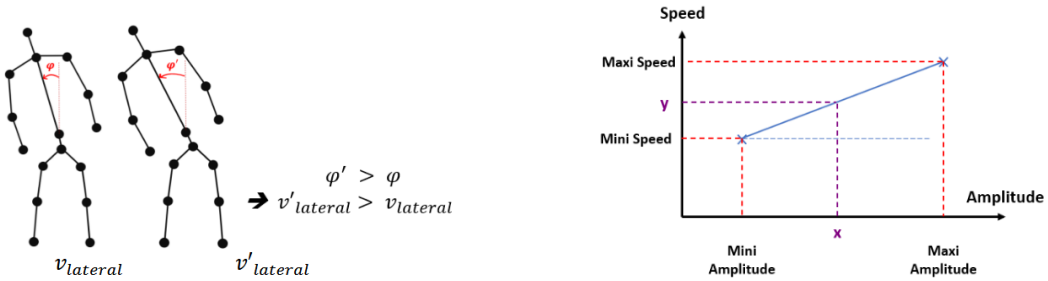
Command	Forward	Backward	Right / Left	Rotation Right / Left	Up / Down
Movement	Raise both arms outwards	Cross arms in front of body	Lean Right / Left	Torso rotation to the Right / Left	Bend Backward / Forward
Body posture					

Figure 2.6: Commands and associated gestures for the Microsoft Kinect system

drones velocity. Larger movements correspond to a higher speed command where as smaller, less pronounced movements result in lower speed commands. This form of control was achieved using a continuous linear mapping between the gesture amplitude and the speed of the gesture. Figure 2.7 (Gio *et al.*, 2021) shows how the amplitude and speed of the gestrure is used to control the speed of the drone.



(a) Diagram showing the control of the spped for a lateral movemnt of the drone.

(b) Graph showing the linear regression for speed calculation

Figure 2.7: Speed control of the Kinect system

Figure 2.7a shows that the angle φ' is greater than φ; which results in the lateral speed $v'_{lateral}$ of the drone being greater than the lateral speed $v_{lateral}$ of the drone. The following formulas are used to caluclate the gradient and the equation for the linear regression line shown in Figure 2.7b:

$$m = \frac{\text{maxi Speed} - \text{mini Speed}}{\text{maxi Amplitude} - \text{mini Amplitude}} \tag{2.1}$$

$$y = (x - \text{mini Amplitude}) \times m + \text{mini Speed} \tag{2.2}$$

Gio *et al.* (2021) peformed a flight test to evaluate the systems peformance. The experiment was designed to reflect realistic future applications and required participants to navigate the drone to a target location while avoiding obstacles, capture an image, and return to the starting point. Both experienced and inexperienced users were included to represent professional and casual use cases. Although live video from the drone's onboard camera was displayed, the test was carried out in an open space to ensure that less experienced users could still maintain direct visual contact with the drone. To enable comparison, the same task was performed using both gesture-based control and a traditional controller. Four participants took part: one experienced with both control methods, one familiar only with conventional control, and two with no prior drone experience. Users were given a brief explanation of the gestures and a 30-second practice period for each method, and each participant completed the task twice to ensure reliable data collection. The results are summarised in Figure 2.8.

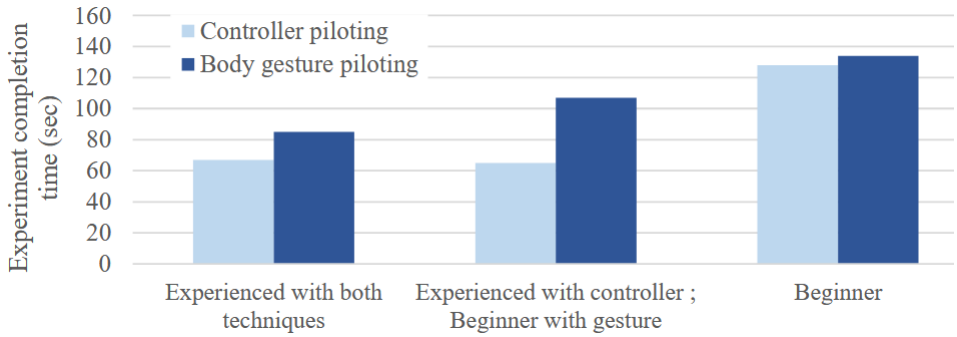


Figure 2.8: Results of the Kinect gesture control experiment

The results demonstrate that the Kinect-based gesture control system is both reliable and effective for drone operation. Gesture recognition was successful for all participants, regardless of physical differences, confirming the robustness of the angle-based tracking approach. The system also showed high accuracy, with less than 2 % error between intended and executed speed commands, indicating precise translation of gestures into control actions. In practical flight tests, all users—including beginners, were able to complete navigation tasks successfully, although gesture-based control was about 27 % slower for experienced users compared to a traditional controller. Despite this, the system was found to be more intuitive and less mentally demanding, particularly for inexperienced users, highlighting its potential as a natural and user-friendly alternative to conventional drone control.

2.2.5 Googles MediaPipe Hands

A widely used image-based gesture control system in research is Google's MediaPipe Hands (MPH) framework. Khaksar *et al.* (2023) has used this software to develop a quadcopter control system as an alternative Human-Computer Inteaction (HCI) approach, focusing on replacing conventional drone control methods with a Hand-Gestuer Recognition (HGR) interface. The aim of this system was to develope and evaluate a HGR system capable of providing an intuative, low-cost, and reliable means of controlling a quatcopter without the neeed for specialised controllers or wearable hardware.

The cental element of this system is the MPH framework. The authors descibe the MPH framework as an on-device real-time hand identification software, designed to operate using a single RGB camera. MPH operates using a two-stage machine-learning pipeline: first, a palm detection model identifies regions of interest within an RGB image, and second, a hand landmark model estimates a 21-point skeletal representation of the detected hand. Figure 2.9 shows the landmarks of the hand (Taylor *et al.*, 2025).

These landmarks provode a 2.5D representation (inferring depth) that allows for

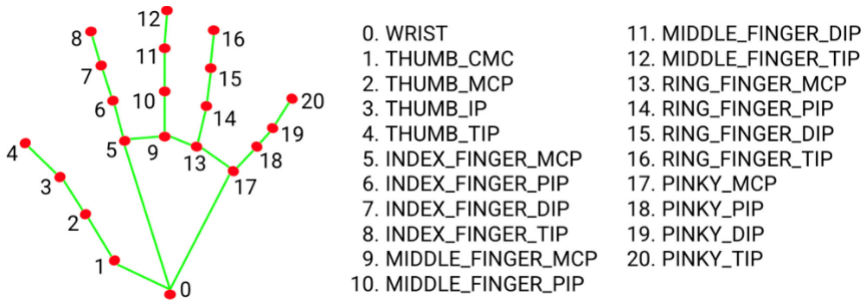


Figure 2.9: Landmarks identified by MediaPipe Hands

the calculation of joint angles and spatial relationships between the fingers. A quantitative validation was performed by comparing these computed joint angles with clinically measured values using a goniometer, allowing the authors to assess model accuracy under different hand poses and orientations. Finally, multiple classification algorithms are tested to determine the most accurate and efficient method for mapping gestures to predefined drone commands, and the system was implemented and tuned through real-world flight tests.

However, a limitation of MPH is that it is sensative to depth (Z-axis) variations (Khaksar *et al.*, 2023), wich could reduce the landmark accuracy under certain conditions. It was found that the landmark accuaracy decreased from 86.7% to 41.5% under certain conditions. Despite these limitations, the system authors proposed a classification method that greatly improved the accuracy and allowed for the to maintain reliable performance achiving a gesture-calssification accuracy of 96.25 % in real-world flight tests.

Additionally, Taylor *et al.* (2025) has built on the use of MediaPipe Hands for gesture control of drones. Making use of the 21 landmaks, the authors were able to implement a custom function that extracts two types of featires from the landmark data. These include:

- **Distance features**: The eclidean distacne between specific landmarks were calculated as shown in Equation 2.3.

$$d_{ij} = \left|v_i - v_j\right| = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \tag{2.3}$$

- **Angle features**: θ_{ijk} was defined using the cosine rule (Equation 2.4) to measue the geometric angles between three consecutive landmarks, for example the finger joints.

$$\theta_{ijk} = \arccos\left(\frac{(v_i - v_j)^T (v_k - v_j)}{d_{ij} \cdot d_{kj}}\right) \tag{2.4}$$

Using these equations, a gesture matrix is constructed, mapping each equation output to a specific gesture command, as illustrated in Figure 2.10 (Taylor *et al.*, 2025).

$$G = \begin{cases} G_{pointing}, & if \quad \theta_1 > 120^\circ\ \theta_5 > 120^\circ\ \theta_6 > 100^\circ\ \theta_7 > 100^\circ\ \theta_8 > 120^\circ \\ G_{peace}, & if \quad \theta_0 > 120^\circ\ \theta_5 > 120^\circ\ \theta_1 > 120^\circ\ \theta_6 > 120^\circ\ \theta_7 > 100^\circ\ \theta_8 > 100^\circ \\ G_{thumbsUp}, & if \quad \theta_4 > 100^\circ\ \theta_6 > 100^\circ\ \theta_7 > 100^\circ\ \theta_8 > 100^\circ\ d_0 > 225 \\ G_{pinky}, & if \quad \theta_3 > 120^\circ\ \theta_5 > 130^\circ\ \theta_6 > 130^\circ\ \theta_7 > 130^\circ\ \theta_8 > 120^\circ \\ G_{openHand}, & if \quad \theta_0 > 120^\circ\ \theta_5 > 120^\circ\ \theta_6 > 120^\circ\ \theta_7 > 120^\circ\ \theta_8 > 120^\circ \\ G_{fist}, & if \quad \theta_4 > 100^\circ\ \theta_6 > 120^\circ\ \theta_7 > 100^\circ\ \theta_8 > 100^\circ\ d_0 < 225 \\ G_{okay}, & if \quad \theta_4 < 100^\circ\ \theta_1 > 120^\circ\ \theta_2 > 120^\circ\ \theta_3 > 120^\circ\ \theta_6 > 120^\circ\ \theta_7 > 120^\circ\ \theta_8 > 120^\circ \\ G_{love}, & if \quad \theta_0 > 120^\circ\ \theta_3 > 120^\circ\ \theta_5 > 120^\circ\ \theta_6 < 130^\circ\ \theta_7 < 130^\circ\ \theta_8 > 120^\circ \\ G_{other}, & \quad otherwise \end{cases}$$

Figure 2.10: Gesture matrix defining specific gestures based on distance and angle features extracted from the MediaPipe Hands landmarks.

The results of this system showed that 8 of the 10 gestures reached a 100 % recognition rate, however there were minor inconsistancies with the "peace" gesture achieving an accuracy of 88 % and the "thumbs up" gesture achieving an accuracy of 96 % as these appeared similar to other gestures. Thus the researchers concluded that the system successfully bridges the gap between human interaction and autonomous safety. While this system was shown to be effective, the researchers identified that the lighting conditions, hand positioning, and computational overhead were the primary areas for future optimization.

2.2.6 DJI Neo 2

The most recent and advanced commercially available drone that uses gesture recognition to control its movement is the DJI Neo 2. This drone was released in November 2025 (Kieldsen, 2025). Due to this drone being very new there is little information available about the specific gesture recognition system that it uses, however the unique features of the drone are known. They include:

- Palm takeoff and landing
- Distance control
- Position Control

According to the user manual (DJI, 2026), the drone can be controlled through gestures. When powered on, the user must raise their palm towards the camera, the drone will then indicate with a blue light that the drone can be controlled with gestures. The drone then tracks the position of the user's hand and can move in the direction of the hand movement. For example, if the user moves their hand to the

left, the drone will move to the left. If the user moves their hand up, the drone will move up. This is decsribed in Figure 2.11.

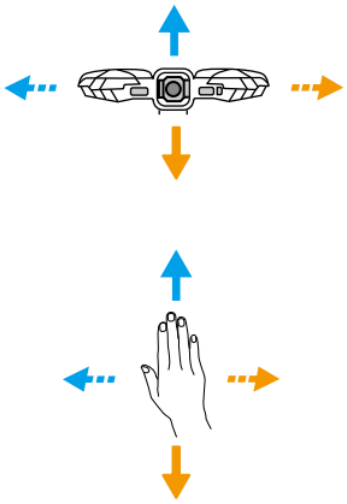


Figure 2.11: Position control of the DJI Neo 2 using hand gestures.

The drone can also be controlled by moving the hand closer or further away from the camera, which controls the distance of the drone from the user. This is done by moving both hands closer together to bring the drone closer, and moving them further apart to move it further away as seen in Figure 2.12 (DJI, 2026).

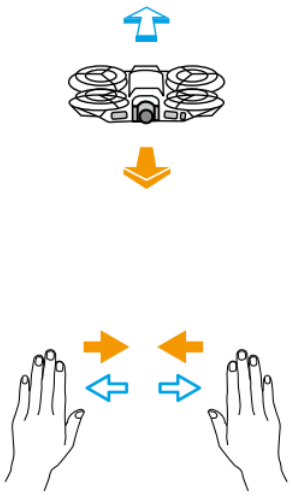


Figure 2.12: Distance control of the DJI Neo 2 using hand gestures.

Additionally, the DJI drone has other features that are more specific to photography and videography. For example the drone allows the user to select *ActiveTrack*, *Spot-*

light and *Quickshots*. The Quickshots feature on DJI drones are specific cinematic shots that have a set flight path. They are:

- Dronie
- Helix
- Rocket
- Circle
- Boomearang

In order to make use of these features, the user needs to use the buttons on the side of the drone to select the specific feature they wish to use.

A notable limitation of this system is that gesture control is restricted to positioning the drone, with all additional functions requiring manual configuration on the drone itself or via a remote controller. This project therefore aims to address this limitation by developing a system capable of interpreting more complex gestural commands, enabling functions such as capturing photos, recording video, and activating specific flight modes to be done with a hand gesture.

2.3 Gesture Classification Techniques

Having discussed the different types of gesture control and the current implementations of gesture control of drones, it is also important to discuss the different techniques that are used to recognise and classify gestures. There are a number of different techniques that have been used and the advancement of machine learnign has allowed for more accurate and efficient gesture recognition software.

2.3.1 Traditional Methods

In this report, traditional methods refer to the well established and commonly used techniques that are used to recognise and classify gestures. Traditional hand gesture recognition methods rely on manually designed features extracted from sensor data, combined with classical machine learning classifiers. For vision-based systems, common features include contour-based descriptors, Zernike moments, and Histograms of Oriented Gradients (HOG) (Bhushan *et al.*, 2022). For signal-based systems using wearable devices or sensors, statistical measures such as Variance (VAR), Root Mean Square (RMS), and Mean Absolute Value (MAV) are used to characterize the signal (Nogales and Benalcázar, 2023). These extracted features are then passed to classifiers such as K-Nearest Neighbors (KNN), Support Vector Machines (SVM), Naive Bayes, or Decision Trees (Bhushan *et al.*, 2022).

However, traditional methods suffer from significant limitations that restrict their real-world applicability. Vision-based approaches are highly sensitive to lighting conditions and occlusions, which degrade feature quality and reduce system robustness. For dynamic gestures, models such as Hidden Markov Models (HMMs) and Dynamic Time Warping (DTW) struggle to capture complex temporal patterns, while also requiring subjective parameter tuning that often fails to generalize across users (Jiashan and Zhonghua, 2024). More fundamentally, manually designed features suffer from the curse of dimensionality—too many features lead to overfitting—and systems cannot adapt to natural variations in how gestures are performed across different users or contexts (Jiashan and Zhonghua, 2024). These limitations prompted the shift toward automated feature learning through deep learning approaches.

2.4 New Advancements

In response to the limitations of traditional gesture recognition methods, there has been significant development in the field of gesture control. These advancements extend beyond drone applications, enabling new forms of human–robot interaction and improved recognition and classification techniques. The most relevant developments are discussed below.

2.4.1 ToF Camera in Moving Vehicle

Reyes *et al.* (2023) developed a system that uses a Time of Flight (ToF) camera to capture hand gestures in a moving vehicle. The aim of this research was to adress the challenges of dynamic hand gesture recognition within a driving environment. The researchers propose a model-adaptation approach to personalize a CNNLSTM (Convolutional Long-Short Term Memory) neural network, aiming to improve recognition accuracy while reducing the massive data requirements typically associated with deep learning. The system that was developed allowed a driver of a vehicle to control the infotainment system of the car using gestures to peform task such as increasing the volume, or skipping a song. It is important to note that the authors state that the system is designed to be expanded upon and adapted to be used in other applications.

A ToF camera was chosen due to the limitations of RGB cameras in environments with variable lighting conditions. The data aquisition process involed the following:

- **Camera setup**. The camera was mounted on the ceilign of the vehicel near the front of the car. This provided a top-down perspective of the gear shift area.
- **Data collection**. 83 participants were asked to perform a series of six specified

gestures. These gestures can be seen in Figure 2.13 (Reyes *et al.*, 2023). As a result the final dataset comprised of 4325 gestures.

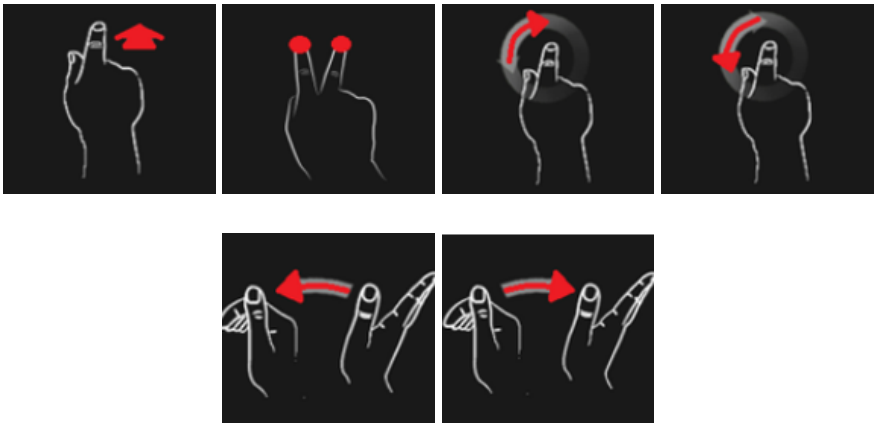


Figure 2.13: Six gestures performed by participants

The dataset then underwent a pre-processingng phase which included stadarising the data such that each recorded gesture was fiixed to 12 fps (frames per second), having a maximum of 70 frames per gesture. Secondly a 3D region of interest was established to be within 12 - 500 cm from the camera to limit where a gestre could be peformed. Finally background subtraction was used to isolate the hand from background of the car. With this completed the data was then used to train a CNNLSTM model.

The study highlight that a "one-model-fits-all" approach is insufficient due to individual differences in gesture performance. The model personalization and adaptation process begins with the creation of a Universal Background Model (UBM), a baseline CNNLSTM architecture trained on a broad group of participants that initially achieves a recognition accuracy of 66.9 %. To account for the individual differences in how various users perform dynamic gestures, the researchers utilize fine-tuning to adapt this UBM to specific individuals within an "Adaptation Set". This single-step adaptation involves training the UBM for an additional 20 epochs using a small amount of data from the specific user, which increases recognition accuracy to 77.8 %. The effectiveness of this personalization is further enhanced through data augmentation, using random translations along the X and Y axes to artificially expand the training set, which pushes the average personalized accuracy to 86.4 %. Experimental results indicate that while the model improves with as few as two gestures per class, accuracy peaks at 90.5 % when the model is fine-tuned with 20 gestures per class. Finally, the study explores incremental learning, where the vehicle adapts to the driver over time; however, this results in a "stability-plasticity dilemma" where the system becomes highly accurate for the specific user but "forgets" how to recognize gestures from others, a common side effect of deep learning

personalization.

2.4.2 One-shot Gesture Recognition Framework

Cabrera *et al.* (2017) identified a core problem with gesture recognition systems, that being, they require many training examples to work. As a result the researchers developed a system that learns to recognise a gesture from a single observation. This is known as one-shot learning. Rather than collecting lots of training data and treating the data as purely statistical patterns, this framework extracts key "inflection points" from a single gesture and uses them to artificially generate hundreds of realistic variations that are used for training (Cabrera *et al.*, 2017).

The inflection points used in the model are not arbitrary; they are selected to reflect meaningful changes in gesture motion. Their choice is biologically motivated and aligns with the way humans represent gestures in the motor cortex, as reflected by electroencephalographic (EEG) signals.

The approach was evaluated using eight upper-limb gestures from the MSRC-12 Kinect dataset, across multiple classification techniques, including Dynamic Time Warping (DTW), Support Vector Machines (SVM), Hidden Markov Models (HMM), and Conditional Random Fields (CRF). Results demonstrated strong performance, achieving an average classification accuracy of approximately 90 % across all classifiers. In addition to conventional accuracy metrics, the authors introduced a novel “coherency” measure, which evaluates the agreement between human and machine interpretation of gestures. Using a Baxter robot to perform the generated gestures, a coherency score of 93.8 % was achieved, indicating a high level of alignment between artificial and human gesture recognition.

Furthermore, the system exhibits adaptive learning capabilities, where performance improves as additional real-world observations are incorporated. Accuracy was shown to increase from approximately 90 % in the one-shot case to around 95 % with ten training examples, illustrating a natural progression from one-shot to N-shot learning. However, a temporary decrease in performance at low sample sizes highlights the sensitivity of the system to outlier observations. This can be seen in Figure 2.14 which shows the Adative Learning (AL) system implemented with two differrent classification algorithems: DTW and HMM.

Overall, the key contribution of this work lies in its classifier-agnostic framework, which integrates principles from cognitive science with machine learning to enable robust recognition of previously unseen gestures from minimal training data.

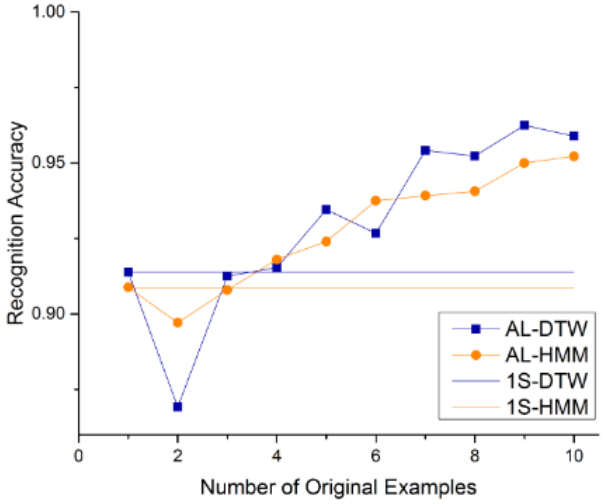


Figure 2.14: Adaptive learning performance of the one-shot gesture recognition system, showing classification accuracy as a function of the number of training examples for both DTW and HMM classifiers.

2.4.3 Few-Shot Learning Vision-based Hand Gesture Recognition

Following on from this research, Shahi *et al.* (2024) developed a system that also learns to recognise a gesture from a single observation. The authros of this research present a vision-based hand gesture recognition system that enables gesture customization from a single demonstaration. The system proposed by Shahi *et al.* (2024) developed a end-to-end framework that supports a wide range of gestures, including both static and dynamic, one and two-handed gestures as well as gestures from multiple viewpoints. The method combines hand keypoint extraction, a graph transformer architecture, and a few-shot learning (FSL) stratagy besed on model-agnostic meta-learning (MAML) to rapidly adapt to new gesture classes from minimal data. Hand pose estimation is used to extract skeletal joint coordinates, which are processed by a graph transformer to capture both the spatial relatiobships between the joints and the temporal motion patterns. To further enhance learning from limited data, a meta-augmentation technique is introduced to increase task diversity by generating a large number of additional gesture combinations to build a greater dataset. Unlike many prior works, the system also incorporates a background class to distinguish intential gestures from irrelevant hand movements, improving real world robustness. Figure 2.15 gives a good overview of the system (Shahi *et al.*, 2024).

Shahi *et al.* (2024) evaluated the system through a user study involving 21 participants and 20 gesture classes. The results showed an average accuracy of 95 % with only one demonstration per gesture. This shows that the system outperforms

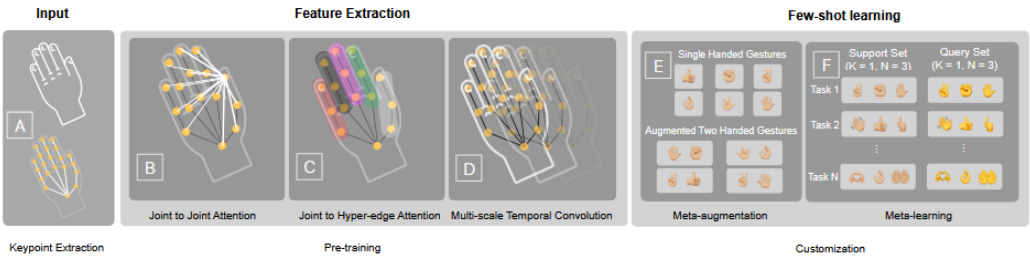


Figure 2.15: Overview of the vision-based few-shot hand gesture recognition system.

baseline approaches such as fine-tuning, nearest neighbours and random forest classifiers. Additionally, the authors demonstrate the practicallity of their approach via three real-wolrd applications, including a gesture design tool, gesture-based productivity interfaces, and mixed reality interaction systems.

2.4.4 Neural Bands

One final advancement worth noting is that use of neural bands. These neural bands are made up of electromyography (EMG) sensors that are mounted on a wrist and are used to sense the electrical signals of a persons muscles.

Kaifosh and Reardon (2025) developed a wristband-based system that translates muscle electrical signals into computer commands, offering a hands-free input method that does not require any device to be held or touched. Unlike existing high-performance brain-computer interfaces, which require invasive brain surgery and are calibrated specifically for each individual user, this system is designed to work across a broad population without user-specific setup. The wristband captures electrical signals produced by the muscles of the forearm and wrist, which reflect the movement commands sent by the brain, and converts these into computer inputs in real time. Figure 2.16 show the design of the nerual band (Kaifosh and Reardon, 2025).

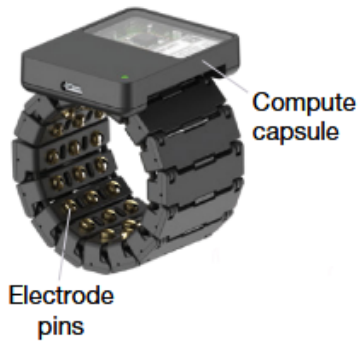


Figure 2.16: Neural Band Design

A key achievement of the study is that the system generalises across users without individual calibration. This was made possible by collecting training data from thousands of participants and using this data to train neural network models capable of recognising movement patterns across diverse individuals. The authors found that model performance improved predictably as more training data and larger models were used, a pattern consistent with findings in other large-scale machine learning domains such as language modelling.

The system was tested in live use scenarios, where new users with no prior experience achieved handwriting speeds of 20.9 words per minute and could reliably detect discrete finger gestures at a rate of 0.88 per second. The authors also showed that a brief 20-minute personalisation session could improve handwriting accuracy by a further 16 %, particularly for users who initially struggled with the generic model. The authors suggest this type of interface could benefit both everyday mobile computing and assistive technology applications for individuals with limited motor function.

A numebr of companies have since developed similar products, most notably, Meta (Meta, 2026) and Mudra Wearables Inc (Mudra Wearable Inc., 2026). The neural band developed by Meta is used to allow users to control the Meta Ray-Band display as well as the Orion ecosystems (Meta, 2026). On the other hand, the Mudra band was designed for a more general perpose. This band allows users to control a wide range of devices such as computers and smart TV's, and can eliminate the need for a mouse, keyboard or touchscreen (Mudra Wearable Inc., 2026). Both the Meta band and the Mudra band can be seen in Figure 2.17.

(a) Meta's Neural Band (b) Mudra's Neural Band

Figure 2.17: Currently Available Commercial Neural Bands

Chapter 3

Workplan

This section outlines the proposed work plan for the project, detailing the key phases and activities required to achieve the stated objectives. The plan follows a structured approach, progressing from foundational research through system design, prototype development, testing, and final documentation.

- **Research**
  A literature review will be conducted to develop an understanding of gesture control, its underlying principles, and its various applications. This review will help clarify the problem space, enabling a well-defined problem statement. It will also provide a basis for comparing existing gesture recognition techniques and implementations.
- **Concept Definition**
  - **Aim and Objectives**
    This section will clearly define the overall aim of the project, along with the specific objectives required to achieve it.
  - **Stakeholders' Requirements**
    This will identify the needs and expectations of the end users. It will define who the system is intended for and outline the key functionalities and performance expectations from a user perspective.
  - **Engineering Requirements**
    The technical requirements of the system will be established here, guided by the stakeholders' needs. These requirements will translate user expectations into measurable engineering specifications.
  - **Drone Selection**
    A comparison of potential drones will be conducted, highlighting the advantages and limitations of each option. This will support the selection of the most suitable platform for the project.
- **System Architecture Design**
  - **Model Selection**
    A suitable gesture recognition and classification approach will be selected. This includes evaluating existing methods and identifying opportunities for improvement.

  - **Model Development**
    The chosen model will be developed, including training procedures and parameter tuning.
  - **Model Optimisation**
    The model will be refined to improve performance, accuracy, and computational efficiency.

- **Prototype**
  - **Hardware Selection**
    Appropriate hardware components, such as cameras and sensors, will be selected. The drone selection is treated separately, as it significantly influences other hardware choices.
  - **Assembly**
    The system components will be assembled into a functional prototype, including the final test drone setup.
  - **Hardware Testing**
    The hardware will be tested to ensure reliable operation and to identify and resolve any faults.
  - **Data Collection Protocol**
    A structured method will be designed for safely collecting and storing data from both the drone and the gesture recognition system.
  - **Software Implementation**
    Once the hardware is operational and data collection methods are in place, the software will be integrated with the hardware, and initial system testing will begin.
  - **Software Evaluation**
    System performance will be evaluated, and improvements will be made based on the results.
  - **System Implementation**
    The fully integrated system will be tested in various environments and across different use-case scenarios to assess robustness and adaptability.
  - **Ethics Approval**
    Ethical approval will be obtained, as the system testing involves human participants.
  - **User Testing**
    Participants will be asked to perform a range of gestures to evaluate how effectively the system responds to different users and gesture variations.

  - **Results Analysis**
    The collected data will be analysed to assess system performance and identify areas for improvement.

- **Documentation**
  The project report will be continuously developed throughout the project. This ensures that all findings, results, and insights are accurately recorded and preserved.

Chapter 4

Project Schedule

This section provides an overview of the planned activities and timeline for the project.

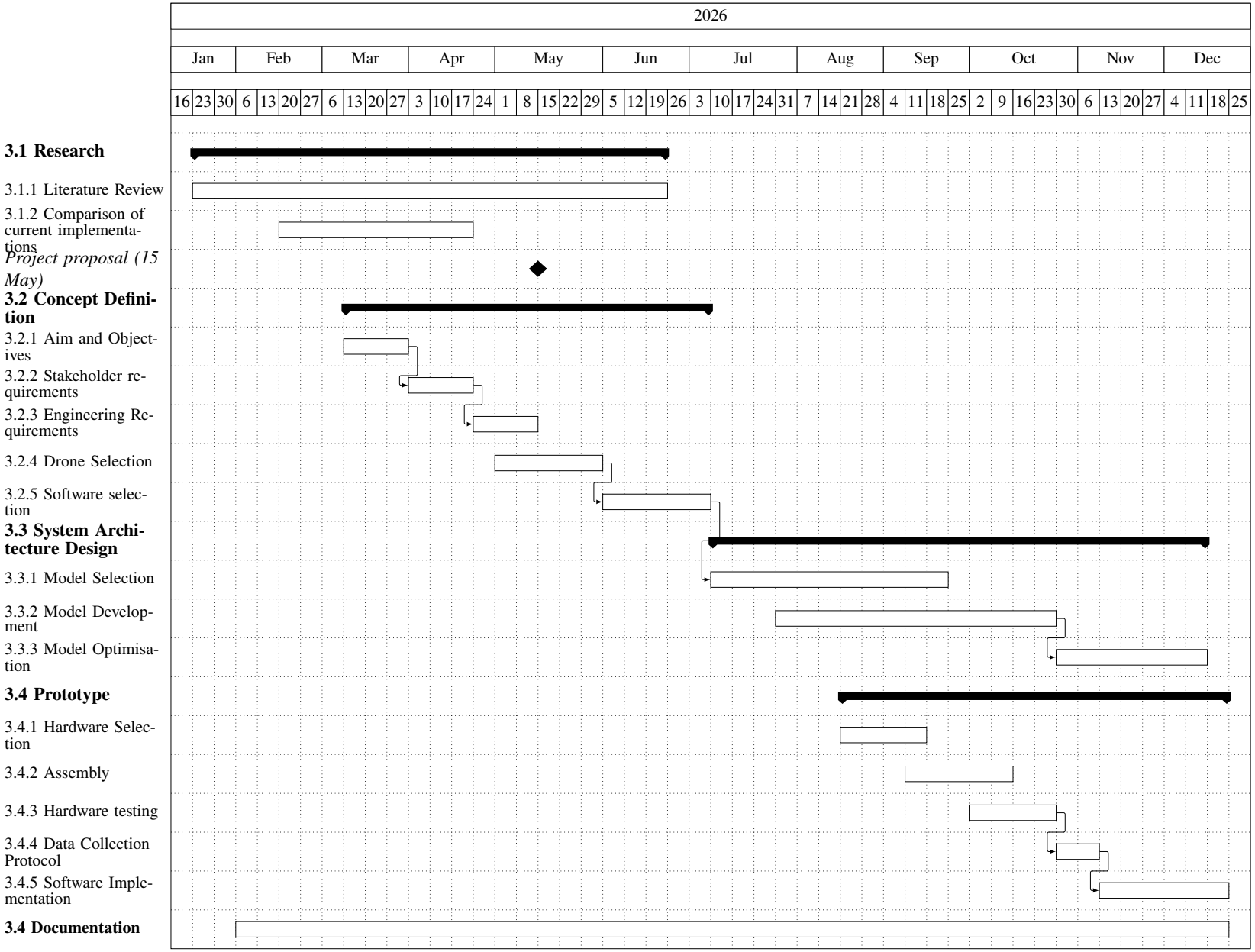


Figure 4.1: Gantt Chart 2026

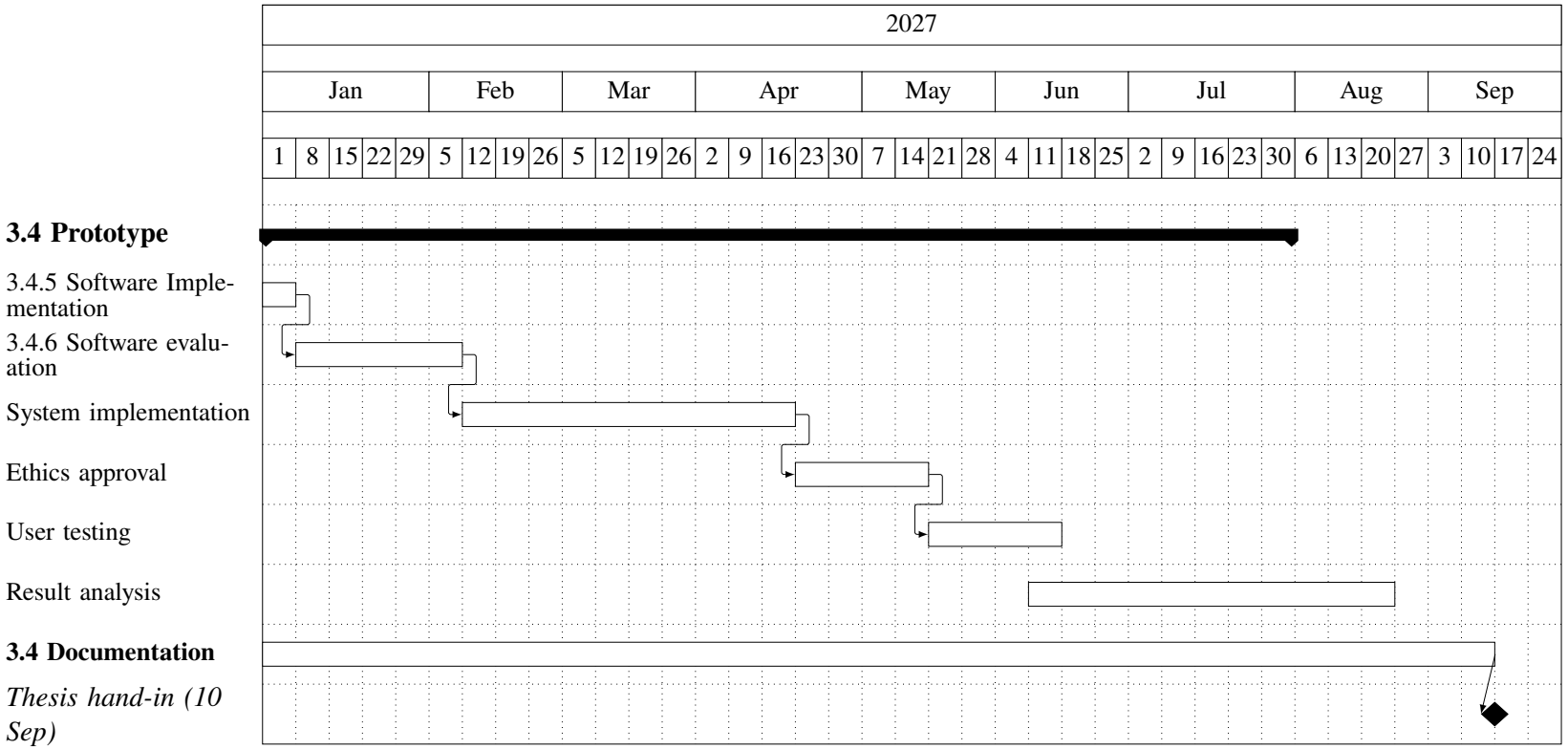


Figure 4.2: Gantt Chart 2027

Chapter 5

Risk Assessment

This section presents a risk assessment for the development, assembly, and testing of the gesture-controlled drone system. Potential hazards associated with each activity are identified and classified according to their impact on personal safety (P), equipment (E), or both. Appropriate mitigation strategies are proposed to reduce the likelihood and severity of each risk. The assessment follows a proactive approach, ensuring that safety, regulatory compliance, and system reliability are considered throughout all stages of the project.

Table 5.1: Risk assessment for gesture-controlled drone system.

Activity	**Risk**	**Risk Type* (P/E)**	**Mitigation**	**Risk Severity**
General				
Ethics approval	Delay in approval prevents user testing	E	Submit ethics application early, ideally during concept phase	High
Flight operations	Flying without permits and breaches of regulations	P & E	Identify regulations and obtain permits before flight	High
Hardware procurement	Delays in components delay entire project	E	Order early and identify backup suppliers. Additionally, purchase spare parts to account for faulty or damaged components	High
Battery storage	Fire risk from incorrect LiPo storage	P & E	Store in LiPo bag and use approved chargers. Have an appropriate fire extinguisher near in the labs where the batteries are stored	High
Software				
Gesture control	Misclassification causes unintended motion	P & E	Implement kill switch and test in simulation first	High

Continued on next page

Activity	**Risk**	**Risk Type* (P/E)**	**Mitigation**	**Risk Severity**
Model optimisation	Excessive tuning delays project	E	Define stopping criteria and time-box optimisation	Medium
Data collection	Data loss or corruption during testing	E	Backup data regularly and save to multiple devices	Medium
Assembly				
Soldering	Burns or inhalation of fumes	P	Work in ventilated area and use proper handling	Low
Electronics handling	Damage from electrostatic discharge or overcharge	E	Ensure all components are grounded properly and power components at rated voltage and current settings	Low
Electrical connections	Shock from live wiring	P	Disconnect power before modifications	Medium
Battery handling	Fire from damaged LiPo battery	P & E	Inspect batteries and avoid short circuits. Have quick access to appropriate fire extinguisher	High
Drone assembly	Cuts from sharp edges	P	Where appropriate personal protective equipment and sand sharp edges	Low
Tool use	Injury from slipping tools	P	Use correct tools and secure workpiece. Get appropriate training if needed	Low
Workbench	Accidents due to clutter	P & E	Maintain clean and organised workspace	Low
Testing				
Initial flight tests	Collision with people or objects	P & E	Use safety zone, prop guards, and spotter	High
User testing	Injury due to incorrect gesture response	P & E	Maintain safe distance and manual override	High
Outdoor testing	Loss of control (flyaway)	P & E	Use RC override and test in enclosed areas	High
Environmental conditions	Wind or obstacles cause collision	P	Take precautions to plan tests to mitigate environmental interference.	Medium
Pre-flight checks	Battery failure causes crash	P	Inspect and monitor batteries before flight	High

Continued on next page

Activity	**Risk**	**Risk Type* (P/E)**	**Mitigation**	**Risk Severity**
Hardware failure	Loss of vision/processing mid-flight	P & E	Implement watchdog and pre-flight checks	Medium

**P – Personal, E – Equipment*

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